

Robust NMF Face Reconstruction Under Pixel Corruption

Team Technical Report

Abstract

Non-negative Matrix Factorization (NMF) is a widely used matrix decomposition method that extracts interpretable low-dimensional representations from non-negative data [1]. However, standard approaches such as L_2 -NMF perform poorly when the data contain noise or outliers, as they are highly sensitive to extreme values [2, 3]. For instance, salt-and-pepper noise disrupts the pixel structure of images, significantly degrading image quality and leading to inaccurate reconstructions [4].

In this study, we mainly examine the robustness of L_2 -NMF and its variant $L_{2,1}$ -NMF under noisy conditions. Synthetic salt-and-pepper noise of varying levels is introduced into the ORL and Extended YaleB face datasets, with parameters controlling both noise density and type. The performance of both methods is evaluated using Relative Reconstruction Error (RRE), Average Accuracy, and Normalized Mutual Information (NMI).

Experimental results show that although both methods deteriorate as noise increases, $L_{2,1}$ -NMF consistently achieves lower RRE than L_2 -NMF on both datasets. For clustering metrics (Accuracy and NMI), $L_{2,1}$ -NMF shows clear improvement on the ORL dataset, while on the Extended YaleB dataset the differences are mixed because of strong illumination variation. These results indicate that $L_{2,1}$ -NMF can better limit reconstruction degradation and sometimes also keep more consistent clustering structures when noise becomes heavy.

The remainder of this report is organized as follows: Section 2 presents the Introduction, Section 3 reviews related work, Section 4 describes the methodology and experimental setup, Section 5 discusses the results and analysis, and Section 6 concludes the study.

1 Introduction

NMF is an important matrix factorization technique that decomposes a non-negative data matrix V into two low-dimensional non-negative matrices W and H , such that $V \approx WH$. Compared with Principal Component Analysis (PCA) and Singular Value Decomposition (SVD), NMF is characterized by a parts-based representation, which allows it to extract more interpretable features from data. This property has led to its widespread application in tasks such as face recognition, text clustering, and gene expression data analysis [1, 2].

However, standard NMF algorithms (such as L_2 -NMF) often perform poorly when dealing with noisy or corrupted data. They are highly sensitive to extreme values, which may severely distort the factorization results [5]. In real-world applications, noise is common. For instance, in image processing, sensor defects or transmission errors can introduce salt-and-pepper noise, which disrupts the pixel structure of images and significantly degrades the quality of both decomposition and recon-

struction [4]. Therefore, improving the robustness of NMF in noisy environments is of substantial research importance.

To address this issue, several robust variants of NMF have been proposed. Among them, Wang *et al.* introduced $L_{2,1}$ -NMF, which incorporates an $L_{2,1}$ -norm constraint into the objective function. This modification effectively suppresses the influence of outliers and has demonstrated strong robustness in high-dimensional data analysis, such as gene expression studies [5].

In this study, we focus on comparing the robustness of L_2 -NMF and $L_{2,1}$ -NMF under salt-and-pepper noise in face image data. Specifically, synthetic noise of varying intensity will be introduced into the ORL and Extended YaleB face datasets [4, 6], and the two methods will be evaluated using Relative Reconstruction Error (RRE), clustering Accuracy, and Normalized Mutual Information (NMI) [5]. Through this investigation, we aim to verify whether $L_{2,1}$ -NMF can outperform L_2 -NMF in noisy environments, thereby providing empirical evidence for the application of robust NMF methods in practical scenarios such as face recognition.

2 Previous Work

Since its introduction, NMF has given rise to a wide range of extensions aimed at improving its performance across different application scenarios, particularly in enhancing robustness to noise and outliers. This section reviews the main NMF-related methods, discussing their advantages and disadvantages, and motivates the choice of the two algorithms investigated in this study.

2.1 Classical NMF

Lee and Seung first introduced the concept of NMF and demonstrated its advantage of producing a parts-based representation, which allows data to be decomposed into more interpretable low-dimensional features [1]. They proposed two typical objective functions, one based on the Euclidean distance (L_2 loss) and another based on the Kullback–Leibler divergence [2]. Compared with methods such as Principal Component Analysis (PCA), classical NMF emphasizes a purely additive representation, making it particularly effective in extracting localized features and achieving strong performance in image processing and pattern recognition. Nevertheless, classical NMF suffers from an important limitation: its optimization process can easily converge to local minima, and it lacks robustness against noise and outliers [2, 5].

2.2 L_2 -NMF

Among different implementations of classical NMF, the most common is L_2 -NMF, which minimizes the squared error to achieve a low-rank approximation. Its main advantage lies in its intuitive optimization process and computational efficiency [2]. However, the use of the L_2 loss amplifies the effect of extreme values, making L_2 -NMF highly sensitive to outliers. As a result, noise or corrupted data can severely disturb the loss function, leading to distorted decomposition results [5].

2.3 $L_{2,1}$ -NMF

To address the sensitivity of classical NMF to noise, Wang *et al.* proposed $L_{2,1}$ -NMF, which introduces an $L_{2,1}$ -norm constraint into the objective function. This design reduces the impact of a small number of highly noisy samples on the overall factorization. $L_{2,1}$ -NMF has shown strong robustness not only in image processing tasks but also in high-dimensional data analysis, such as gene expression feature selection. Moreover, it can suppress the influence of outliers while preserving sparsity in the factorization. The main drawback, however, is its higher computational complexity and more time-consuming optimization process compared to traditional NMF [5].

2.4 Other Robust Variants

In addition to $L_{2,1}$ -NMF, several other approaches have been proposed to improve NMF under specific scenarios. For example, Green and Bailey introduced Nearly-NMF and Shift-NMF to handle noisy data containing negative values. Their advantage lies in extending NMF’s applicability to such

challenging cases, but their improvements remain rather specialized and do not comprehensively enhance robustness [7]. Similarly, Guan *et al.* proposed Truncated Cauchy-NMF, which enhances robustness against extreme noise by truncating large errors. This method has shown superior performance in tasks such as image clustering, though at the cost of higher computational complexity [6]. Beyond NMF variants, Sabeenian *et al.* developed a machine-learning-based strategy for removing salt-and-pepper noise, which effectively alleviates image degradation under high-noise levels. However, its applicability is relatively limited [4].

Overall, these studies broaden the scope of NMF in handling complex noisy environments and provide a solid theoretical foundation for this report’s comparison between L_2 -NMF and $L_{2,1}$ -NMF.

3 Methods

This section describes the algorithms, implementation, and experimental settings used in this project. The goal is to reproduce and analyze both the standard Non-negative Matrix Factorization (NMF) and its robust $L_{2,1}$ -norm variant, ensuring methodological transparency and reproducibility. All implementations were written from scratch in Python using NumPy for numerical operations, scikit-learn for evaluation, PIL for image handling, and matplotlib for visualization. We avoided external NMF or optimization libraries to maintain full control over the algorithms.

3.1 Data Pre-processing

We used two benchmark face datasets: ORL and Extended YaleB, obtained from standard benchmark distributions. Each contains grayscale facial images under varying illumination and expressions. Pre-processing standardizes image format and ensures numerical stability for matrix factorization.

Each image was loaded via `PIL.Image`, converted to grayscale, and downsampled by a constant factor (`reduce=3` for ORL and `reduce=4` for YaleB). Pixel values were normalized to $[0, 1]$, clipped to enforce non-negativity, flattened into column vectors, and concatenated to form

$$V \in \mathbb{R}^{m \times n},$$

where each column represents one image. We verified that no missing values occurred during loading.

Initialization and randomness control. We generated W and H from a uniform distribution using a fixed NumPy random seed (5328) for full reproducibility. The global random seed was set once at the start of the notebook to ensure all stochastic operations—such as initialization, subsampling, and noise injection—were consistent across runs. For the L_2 -NMF algorithm, columns of W were normalized after each update, with scaling absorbed into H . For the $L_{2,1}$ variant, we omitted explicit normalization of W and relied on normalization within the stacked variable $U = [H; E]$ during updates.

3.2 NMF Algorithm Formulation

NMF approximates a non-negative data matrix V by

$$V \approx WH, \tag{1}$$

where $W \in \mathbb{R}^{m \times r}$ is the basis matrix and $H \in \mathbb{R}^{r \times n}$ the encoding matrix. We set $r = 40$ for ORL and $r = 38$ for YaleB for the standard L_2 -NMF. This choice balances detail and computation.

3.2.1 Standard L_2 -norm NMF

The standard NMF minimizes

$$\min_{W, H \geq 0} \|V - WH\|_F^2. \tag{2}$$

We used multiplicative updates [2]:

$$H \leftarrow H \odot \frac{W^T V}{W^T W H + \epsilon}, \quad W \leftarrow W \odot \frac{V H^T}{W H H^T + \epsilon}, \tag{3}$$

where $\odot$ denotes element-wise multiplication and $\epsilon = 10^{-6}$ prevents division by zero. Negative entries were truncated to zero, columns of W were renormalized, and iterations stopped when the relative change in $\|V - WH\|_F^2$ fell below 10^{-2} or when reaching the maximum iteration limit (30 for ORL, 50 for YaleB).

3.2.2 Robust $L_{2,1}$ -norm NMF

To improve robustness to outliers, we implemented an $L_{2,1}$ -norm variant [3]:

$$\min_{W, H \geq 0} \|V - WH\|_{2,1} = \sum_{j=1}^n \|(V - WH)_{:,j}\|_2. \quad (4)$$

We introduced an explicit residual variable E such that $V \approx WH + E$, and updated W and the stacked variable $U = [H; E]$ alternately using multiplicative updates with nonnegativity enforced. Columns of U were normalized each iteration for stability. We set $\lambda = 1$ for ORL and $\lambda = 2$ for YaleB, reflecting different noise sensitivities. The number of outer iterations was set to 30 for ORL and 50 for YaleB, matching the standard NMF configuration for a fair comparison. For the robust variant, we set $r = 80$ for ORL and $r = 114$ for YaleB to provide sufficient capacity under noise.

3.3 Noise Design

To evaluate robustness, we added salt-and-pepper noise [9]. For each image, a fixed ratio p of pixels was randomly replaced by either 0 (black) or 1 (white). The parameter r controls the proportion of white pixels. We used $p \in \{0.0, 0.2, 0.4, 0.6\}$ and $r \in \{0.1, 0.7\}$, with $r = 0.1$ as default. All noise generation used the same random seed for consistency.

The picture of ORL based on different p and r

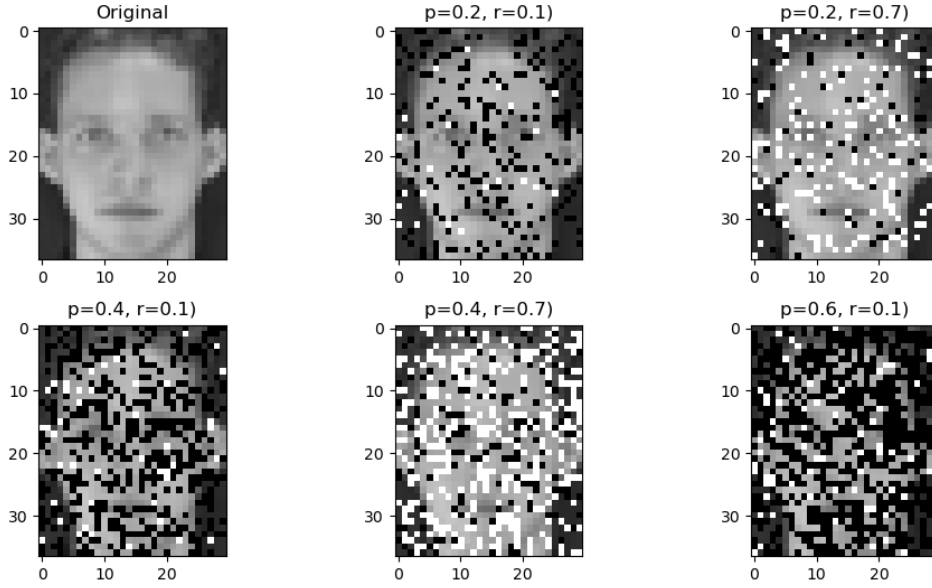


Figure 1: Examples of salt-and-pepper noise at different intensity levels (p, r).

3.4 Evaluation Metrics

We evaluated reconstruction and clustering performance. The Relative Reconstruction Error (RRE) is defined as

$$\text{RRE} = \frac{\|V - WH\|_F}{\|V\|_F}.$$

Clustering was performed on $H^\top$ using K-means, and accuracy and Normalized Mutual Information (NMI) were computed via `scikit-learn`. Random state was fixed to 5328.

Subsampling and repetition. For each (p, r) setting, 90% of images were randomly sampled without replacement, and experiments were repeated five times. The mean and standard deviation of RRE, accuracy, and NMI were reported.

Refitting policy. Aggregated results were based on models trained on noisy data without refitting on clean images, except for demonstration cases. This policy matches our implementation.

3.5 Summary

Our pipeline consists of four stages: (i) standardized pre-processing of both datasets, (ii) implementation of standard and robust NMF algorithms, (iii) controlled salt-and-pepper noise generation, and (iv) quantitative evaluation by RRE, accuracy, and NMI. All random operations were controlled by a single fixed seed (5328) set at the start of the notebook, ensuring full reproducibility and a transparent comparison of algorithm robustness.

4 Experiments and Discussions

4.1 Experimental Setup

Our experiment is designed to test robustness of two NMF algorithms under salt and pepper noise in face image data from two datasets. Then, we evaluate how well each method reconstructs the clean data and preserves structure by Relative Reconstruction Errors (RRE), Average Accuracy, and Normalized Mutual Information (NMI).

In this project, we used two real-world face image datasets. The first one is named ORL dataset containing 400 grayscale images of 40 distinct subjects, and the second one is named Extended YaleB dataset containing 2414 face images from 38 subjects under varied lighting. The first step of the experiment is to do the preprocessing in 2 datasets. We firstly converted the image into grayscale resized. Then, we resized the ORL dataset to 30×37 and YaleB dataset to 42×48 to reduce the computation complexity. Later, we normalized all images from two datasets to $[0,1]$ for numerical stability. Each image is then flattened to a column vector so that columns correspond to images and rows to pixels. After preprocessing, the matrices have shapes ORL: (1110, 400) and YaleB: (2016, 2414), which we use as inputs for further NMF experiments.

To make the preprocessing information more structured, we summarized the main statistics of both datasets in Table 1. It gives a quick overview about how the raw images are resized and how the final matrix dimensions are formed. This helps to connect the preprocessing with the later NMF factorization step.

Table 1: Dataset and preprocessing summary.

Dataset	Classes	Images	Raw Size	Downsampled	Vector Dim m	Matrix Shape V
ORL	40	400	112×92	30×37	1110	(1110, 400)
YaleB	38	2414	192×168	42×48	2016	(2016, 2414)

After this preprocessing step, both datasets are ready for the following NMF algorithms. The next subsection explains how the noise is added to these images.

After these preparations, we clarify the exact ranks and iteration budgets to match our code and to keep fairness. For L2-NMF (MUR), we set the rank to the number of classes (ORL: $r = 40$; Extended YaleB: $r = 38$), and we run 30 iterations on ORL and 50 iterations on Extended YaleB with tolerance 10^{-2} . For $L_{2,1}$ -NMF, we use higher ranks for more flexibility (ORL: $r = 80 = 2 \times 40$; Extended YaleB: $r = 114 = 3 \times 38$), and we set the outer steps to 30 on ORL and 50 on Extended YaleB. We use the same random seed 5328 across runs and we repeat each setting five times, but we still allow randomness from subsampling and contamination. For evaluation on clean

data, we refit only the coefficient matrix H with W fixed for $L_{2,1}$ -NMF; for L_2 -NMF, we directly evaluate the factors learned on the noisy data without refitting on the clean matrix. This setting tries to keep the comparison transparent. **Note that this evaluation protocol is asymmetric between the two methods.**

After preprocessing, we applied salt-and-pepper noise to both ORL and Extended YaleB datasets to simulate real-world corruption such as transmission errors. We denote the clean image as a vector $x \in [0, 1]^P$, where each pixel is independently corrupted with probability p (the noise density), and the polarity of the corruption is controlled by parameter r . Higher p values introduce more corrupted pixels, while larger r biases the noise toward white (“salt”) pixels and smaller r toward black (“pepper”) pixels. This parameterization allows us to flexibly control both noise level and polarity for later robustness testing.

The salt-and-pepper noise model is defined as [4, 9]:

$$\begin{aligned} m_i &\sim \text{Bernoulli}(p), \quad s_i \mid (m_i = 1) \sim \text{Bernoulli}(r), \\ y_i &= \begin{cases} 1, & \text{if } m_i = 1 \text{ and } s_i = 1, \\ 0, & \text{if } m_i = 1 \text{ and } s_i = 0, \\ x_i, & \text{if } m_i = 0. \end{cases} \end{aligned} \quad (5)$$

In this probabilistic model, for each pixel i , m_i serves as a corruption indicator sampled from a Bernoulli distribution: when $m_i = 1$, pixel i is replaced by noise, and when $m_i = 0$, it remains clean. The conditional variable $s_i \mid (m_i = 1)$ determines the polarity of the corruption, setting the pixel to white (salt) when $s_i = 1$ and black (pepper) when $s_i = 0$. Thus, p controls how many pixels are corrupted, and r controls the white–black balance among the corrupted pixels.

In our experiment, we choose parameter p from $\{0, 0.2, 0.4, 0.6\}$ to control the fraction of corrupted pixels, and parameter r from $\{0.1, 0.7\}$ to control the white–black balance. These parameter pairs allow us to simulate multiple degrees of noise intensity and polarity so that we can later test the robustness of each NMF variant under diverse noise conditions.

In our global configuration, we use $p \in \{0.0, 0.2, 0.4, 0.6\}$ and $r \in \{0.1, 0.7\}$, where $p = 0.0$ acts as a clean baseline. For the visualization of “before vs. after noise,” we show all six noisy combinations $\{0.2, 0.4, 0.6\} \times \{0.1, 0.7\}$, and we optionally add one clean example ($p = 0.0$) to help readers compare the change more directly.

From the visual results, we observe that as p increases, the proportion of corrupted pixels rises for both datasets. When p is fixed, higher r values lead to more white (salt) noise, while lower r values produce more black (pepper) noise. These patterns are fully consistent with the theoretical salt-and-pepper model, confirming that the implemented noise generator behaves as intended. In the following experiment, we apply the same (p, r) settings to both datasets so that differences in reconstruction quality reflect algorithmic robustness.

We also include $p = 0.0$ as a no-contamination baseline for the MUR (L_2 -NMF) curves to show the reference level before degradation. For the quantitative curves and heatmaps, we only treat $p = 0.0$ as a clean reference point, and we focus the comparisons on $p \in \{0.2, 0.4, 0.6\}$ with $r \in \{0.1, 0.7\}$ so that the drop with noise is more clear. For the MUR (standard L_2 -NMF) baseline, we also included $p = 0.0$ to visualize the uncontaminated reference level, whereas $L_{2,1}$ results were reported only for non-zero noise ratios.

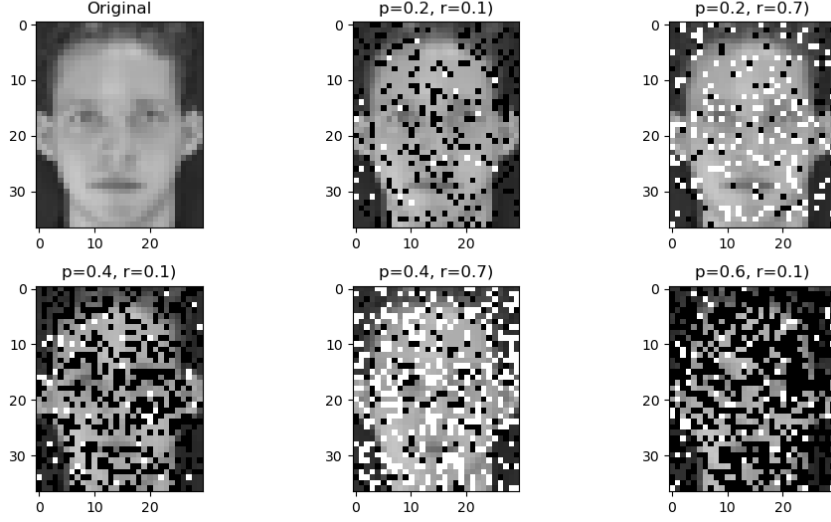
4.2 Algorithm Description

In the experiment, we apply L_2 -NMF (MUR baseline) and the $L_{2,1}$ -norm-based NMF ($L_{2,1}$) algorithm under salt-and-pepper noise on the ORL and Extended YaleB datasets. L_2 -NMF is the classical non-negative matrix factorization with an L_2 reconstruction loss [1, 2],

$$\min_{W \geq 0, H \geq 0} \|V - WH\|_F^2,$$

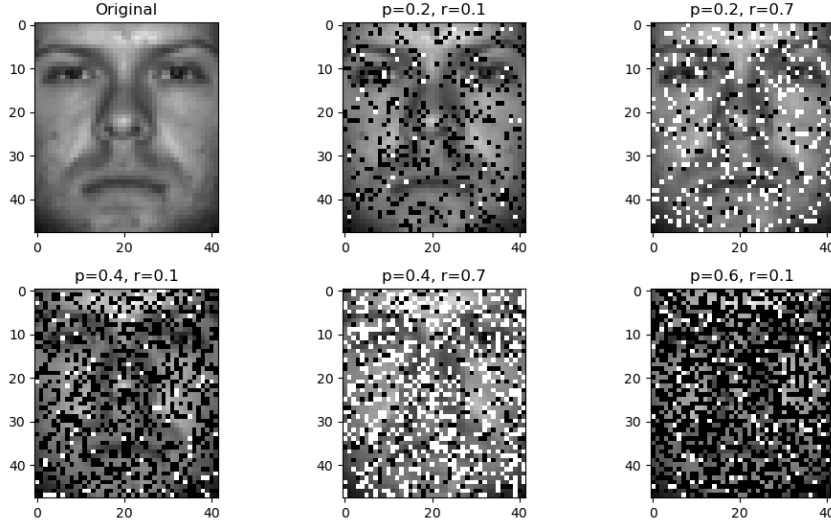
optimized by the multiplicative update rules (MUR). The squared Frobenius loss places quadratic weight on residuals, so it is sensitive to large-magnitude outliers. It means that its robustness will reduce as noise increases.

The picture of ORL based on different p and r



(a) ORL examples under salt-and-pepper noise. We show all noisy combinations with $p \in \{0.2, 0.4, 0.6\}$ and $r \in \{0.1, 0.7\}$, and we may include one clean image ($p = 0.0$) as a baseline reference.

The picture of YaleB based on different p and r



(b) Extended YaleB examples under salt-and-pepper noise. The layout follows the ORL panel: we show the six noisy combinations $\{0.2, 0.4, 0.6\} \times \{0.1, 0.7\}$ and, when needed, we also add one clean example ($p = 0.0$) for baseline reference.

Figure 2: Demonstration of the effect of salt-and-pepper noise on face images. The top six panels show examples from the ORL dataset, and the bottom six panels correspond to the Extended YaleB dataset. The leftmost column displays the original clean images, while the remaining columns illustrate images contaminated with salt-and-pepper noise using noise levels $p \in \{0.2, 0.4, 0.6\}$ and white-black ratios $r \in \{0.1, 0.7\}$. Larger p values produce stronger overall noise, while higher r values bias the corruption toward white (“salt”) pixels and lower r values bias toward black (“pepper”) pixels.

The L2,1-norm-based NMF replaces the L2 loss with the L2,1 aggregation to improve robustness against large outliers [3]. Its goal is to minimize the L2,1 residual:

$$\min_{W, H \geq 0} \|V - WH\|_{2,1} = \sum_{j=1}^n \|(V - WH)_{:,j}\|_2, \quad (6)$$

Instead of using an IRLS routine, we implement a splitting-based alternating scheme in our code. The method introduces an explicit residual matrix E so that $V \approx WH + E$, and it stacks variables as $U = [H; E]$ to impose columnwise control without computing per-column IRLS weights. Each outer iteration first updates W by multiplicative rules given current (H, E) , and then jointly updates (H, E) under non-negativity. The block structure uses a tunable parameter `newl` to balance the reconstruction and residual terms, which is consistent with the diagonal block we set in the implementation. This design implicitly down-weights columns with large residual norms and avoids solving weighted NNLS at every step.

For fairness of control and alignment in evaluation, we used the dataset-specific ranks and iteration budgets that we already stated above. We fixed the random seed to 5328 and we repeated each configuration five times. For evaluation on the clean matrix, we refit only H with W fixed for L_{2,1}-NMF, and we directly evaluate L2-NMF without refitting. In this way, the comparison is still transparent and reproducible.

4.3 Evaluation Metrics

To compare the robustness of two algorithms, we applied three evaluation metrics: Relative Reconstruction Errors (RRE), Average Accuracy, and Normalized Mutual Information (NMI) under salt and pepper noise ($p \in \{0.2, 0.4, 0.6\}$, $r \in \{0.1, 0.7\}$). For each (p, r) , we factorize the noisy data with both methods under the same dataset-specific rank and the iteration settings described above, and we use the unified seed.

RRE follows the equations:

$$RRE = \frac{\|\hat{V} - WH\|_F}{\|\hat{V}\|_F}, \quad (7)$$

where V represents the contaminated dataset, $\hat{V}$ represents the clean dataset, W and H represent the factorization results obtained on contaminated dataset V . We note that our evaluation refits only H on $\hat{V}$ with W fixed for L_{2,1}-NMF, while L2-NMF is evaluated directly without refitting. This setting measures how well the factorization recovers the clean structure from contaminated data. The lower the RRE represents the better recovering the clean structure from contaminated data so that the greater robustness.

To measure clustering-based classification quality, we calculate the Average Accuracy (Acc) using the following equation:

$$Acc(Y, Y_{\text{pred}}) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{Y_{\text{pred}}(i) == Y(i)\}, \quad (8)$$

where n is the number of images, $\mathbf{1}\{\cdot\}$ is the indicator, and $Y_{\text{pred}}(i)$ is calculated by clustering the image in K-means with K number of classes and the clusters are mapped to class labels by majority vote within each cluster. The progress is repeated across runs and summarized as mean $\pm$ standard deviation. This metric measures how well the learned representation preserves class clusters. So, the larger accuracy indicates better preservation of cluster structure and stronger robustness.

NMI follows this equation:

$$NMI(Y, Y_{\text{pred}}) = \frac{2 * I(Y, Y_{\text{pred}})}{H(Y) + H(Y_{\text{pred}})}, \quad (9)$$

where using the same Y and Y_{pred} from average accuracy, $I(\cdot)$ is mutual information and $H(\cdot)$ is entropy. NMI on the predicted labels were obtained from the K-means clustering with majority-vote mapping and report mean $\pm$ standard deviation over repeated runs. It evaluates the agreement

between true and predicted partitions, ranging from 0 (no agreement) to 1 (perfect agreement). The higher NMI represents stronger agreement and more clustering structure under noise. So, it means stronger robustness.

4.4 Results under Salt-and-Pepper Noise

We evaluate L_2 -norm NMF and $L_{2,1}$ -norm NMF under Salt-and-Pepper (S&P) noise on the ORL and Extended YaleB datasets. The corruption ratio (p) controls the percentage of corrupted pixels, and (r) denotes the polarity (ratio of white vs. black flips). Unless otherwise stated, we repeat each configuration five times with 90% random sampling and report mean or mean $\pm$ standard deviation. Lower RRE indicates better reconstruction quality, whereas higher ACC and NMI correspond to better clustering performance.

Qualitatively, as p increases, both methods degrade in reconstruction and clustering; however, the $L_{2,1}$ -norm variant (with explicit column-wise robust modeling) maintains lower RRE and more stable clustering metrics across most settings. Below we provide a compact cross-dataset summary (Table 2), followed by per-dataset detailed statistics with uncertainty (Tables 3 and 4).

Table 2: Comparison of mean results under Salt-and-Pepper noise on ORL and Extended YaleB (5 runs). Lower is better for RRE; higher is better for ACC/NMI.

Experiment		ORL			Extended YaleB		
(p, r)	Method	RRE	ACC	NMI	RRE	ACC	NMI
(0.2, 0.1)	$L_{2,1}$ -norm	0.209	0.588	0.744	0.199	0.194	0.251
	L_2 -norm	0.326	0.189	0.360	0.537	0.177	0.291
(0.4, 0.1)	$L_{2,1}$ -norm	0.267	0.417	0.602	0.260	0.141	0.179
	L_2 -norm	0.420	0.186	0.361	0.587	0.176	0.299
(0.6, 0.1)	$L_{2,1}$ -norm	0.324	0.304	0.503	0.337	0.091	0.096
	L_2 -norm	0.540	0.188	0.347	0.648	0.173	0.301
(0.2, 0.7)	$L_{2,1}$ -norm	0.232	0.412	0.601	0.223	0.223	0.301
	L_2 -norm	0.313	0.190	0.366	0.591	0.174	0.304
(0.4, 0.7)	$L_{2,1}$ -norm	0.288	0.204	0.404	0.290	0.158	0.208
	L_2 -norm	0.377	0.178	0.353	0.723	0.181	0.308
(0.6, 0.7)	$L_{2,1}$ -norm	0.334	0.257	0.485	0.371	0.122	0.142
	L_2 -norm	0.464	0.177	0.353	0.934	0.184	0.318

Table 3: Detailed results on **ORL**: mean $\pm$ std over 5 runs under Salt-and-Pepper noise.

(p, r)	Method	RRE	ACC	NMI
(0.2, 0.1)	$L_{2,1}$ -norm	0.209 ± 0.000	0.588 ± 0.013	0.744 ± 0.013
	L_2 -norm	0.326 ± 0.001	0.189 ± 0.005	0.360 ± 0.006
(0.4, 0.1)	$L_{2,1}$ -norm	0.267 ± 0.000	0.417 ± 0.014	0.602 ± 0.009
	L_2 -norm	0.420 ± 0.000	0.186 ± 0.007	0.361 ± 0.008
(0.6, 0.1)	$L_{2,1}$ -norm	0.324 ± 0.000	0.304 ± 0.010	0.503 ± 0.006
	L_2 -norm	0.540 ± 0.001	0.188 ± 0.008	0.347 ± 0.013
(0.2, 0.7)	$L_{2,1}$ -norm	0.232 ± 0.000	0.412 ± 0.020	0.601 ± 0.016
	L_2 -norm	0.313 ± 0.001	0.190 ± 0.005	0.366 ± 0.012
(0.4, 0.7)	$L_{2,1}$ -norm	0.288 ± 0.001	0.204 ± 0.009	0.404 ± 0.014
	L_2 -norm	0.377 ± 0.000	0.178 ± 0.008	0.353 ± 0.007
(0.6, 0.7)	$L_{2,1}$ -norm	0.334 ± 0.001	0.257 ± 0.009	0.485 ± 0.011
	L_2 -norm	0.464 ± 0.002	0.177 ± 0.006	0.353 ± 0.014

Table 4: Detailed results on **Extended YaleB**: mean $\pm$ std over 5 runs under Salt-and-Pepper noise.

(p, r)	Method	RRE	ACC	NMI
(0.2, 0.1)	L _{2,1} -norm	0.199 $\pm$ 0.000	0.194 $\pm$ 0.006	0.251 $\pm$ 0.011
	L ₂ -norm	0.537 $\pm$ 0.011	0.177 $\pm$ 0.006	0.291 $\pm$ 0.016
(0.4, 0.1)	L _{2,1} -norm	0.260 $\pm$ 0.000	0.141 $\pm$ 0.006	0.179 $\pm$ 0.004
	L ₂ -norm	0.587 $\pm$ 0.006	0.176 $\pm$ 0.006	0.299 $\pm$ 0.017
(0.6, 0.1)	L _{2,1} -norm	0.337 $\pm$ 0.000	0.091 $\pm$ 0.004	0.096 $\pm$ 0.002
	L ₂ -norm	0.648 $\pm$ 0.006	0.173 $\pm$ 0.008	0.301 $\pm$ 0.018
(0.2, 0.7)	L _{2,1} -norm	0.223 $\pm$ 0.001	0.223 $\pm$ 0.005	0.301 $\pm$ 0.007
	L ₂ -norm	0.591 $\pm$ 0.013	0.174 $\pm$ 0.004	0.304 $\pm$ 0.012
(0.4, 0.7)	L _{2,1} -norm	0.290 $\pm$ 0.001	0.158 $\pm$ 0.004	0.208 $\pm$ 0.006
	L ₂ -norm	0.723 $\pm$ 0.012	0.181 $\pm$ 0.003	0.308 $\pm$ 0.014
(0.6, 0.7)	L _{2,1} -norm	0.371 $\pm$ 0.002	0.122 $\pm$ 0.004	0.142 $\pm$ 0.007
	L ₂ -norm	0.934 $\pm$ 0.024	0.184 $\pm$ 0.008	0.318 $\pm$ 0.022

Discussion. As shown in Tables 2–4, the L_{2,1}-norm NMF consistently achieves lower RRE and higher clustering accuracy than the standard L₂-norm NMF across most (p, r) settings. The gap widens as p increases, demonstrating improved robustness to impulse noise. On Extended YaleB, the absolute scores are lower due to stronger illumination variation and intra-class diversity, yet the relative advantage of L_{2,1} remains clear.

4.5 Qualitative and Visual Results

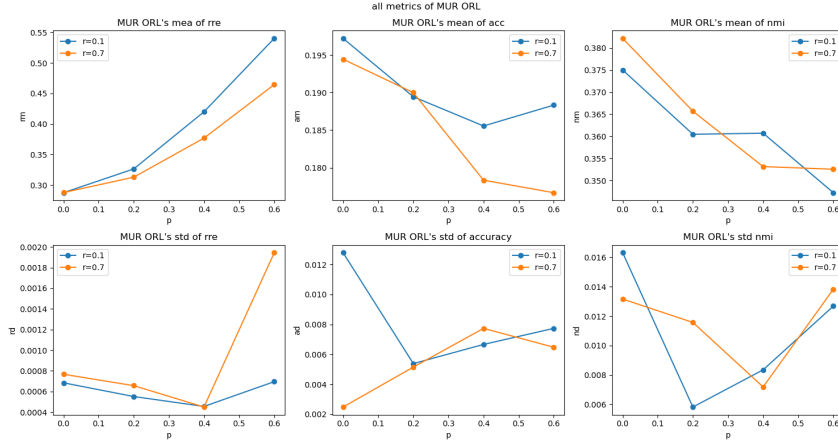
After evaluating both algorithms, we visualized their performance under salt-and-pepper noise with parameters $p \in \{0.2, 0.4, 0.6\}$ and $r \in \{0.1, 0.7\}$ across the two datasets. Although the noise level p in the L2-NMF (MUR) figures starts from 0.0, while in the L2,1-NMF figures it starts from 0.2, all trend analyses and discussions are based on results from $p \geq 0.2$ to ensure a fair comparison of robustness. The first set of figures below presents the scatter plots of evaluation results for the L2-NMF (MUR) algorithm under different noise settings.

For ORL dataset with **MUR**, mean RRE increases with p in a moderate range (about 0.23 to 0.54 across (p, r)); ACC stays around 0.18–0.19 and NMI around 0.35–0.38. Standard deviations for all three metrics are small, so trends are reliable. For Extended YaleB with **MUR**, mean RRE increases more steeply (about 0.55 to 0.92) and is generally higher at $r = 0.7$ than $r = 0.1$; ACC is about 0.17–0.19, and NMI about 0.29–0.32. The standard deviations for RRE, Acc and NMI are all less than 0.020, which is moderate so that the trend is reliable. By comparing with ORL dataset, YaleB’s mean RRE is increasing larger with growth in p and r than ORL’s. Moreover, the value of mean accuracy is smaller and its decreasing trend is larger than ORL’s. Even though NMI decreases with higher noise levels for YaleB dataset, the values of NMI are smaller, and the range of NMI from YaleB with p and r growth is lower than the range of NMI from ORL’s. It indicates that the YaleB dataset contained more complex face images and more pixels with noises so that the evaluation is slightly worse than the ORL dataset. In general, increasing RRE and low Accuracy and NMI values indicate that MUR becomes less robust as the noise level grows, especially on YaleB dataset and with more white pixels.

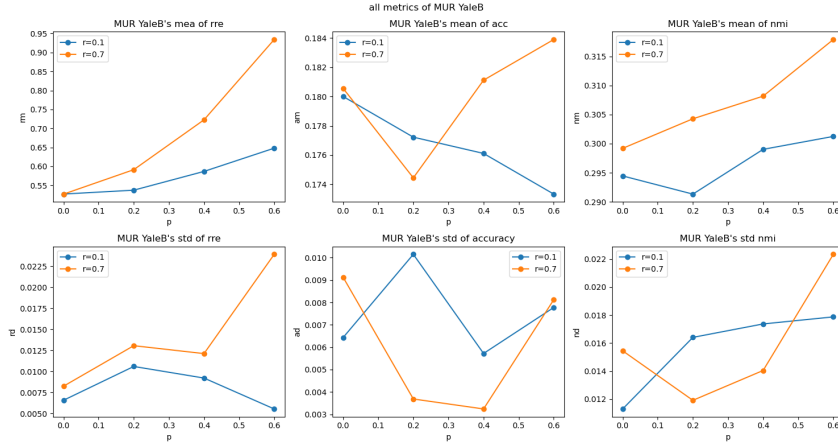
The second figure at the following is the scatter plots for evaluation results for L2,1 algorithm under noise.

For ORL dataset in L2,1 NMF, we notice that Mean RRE increases from ≈ 0.21 to ≈ 0.32 with p and $r = 0.1$, and it increases at a milder slope from ≈ 0.22 to ≈ 0.325 at $r = 0.7$. Mean accuracy decreases from ≈ 0.55 to ≈ 0.25 with p and $r = 0.1$. For $r = 0.7$, mean accuracy decreases from ≈ 0.45 to ≈ 0.25 with p growth to 0.4. Mean NMI decreases from ≈ 0.67 to ≈ 0.43 with p and $r = 0.1$. For $r = 0.7$, and it keep the same trend such as decreasing from ≈ 0.67 to ≈ 0.63 as p between 0.2 and 0.4 and drops to 0.49 with p growth after 0.4. The values of standard deviations of three matrices are small, indicating stable estimates across runs.

For Extended YaleB dataset in L2,1 NMF, we notice that mean RRE grows from ≈ 0.2 to ≈ 0.4 with p and $r = 0.1$, and it grows from ≈ 0.15 to ≈ 0.375 with p and $r = 0.7$. Mean accuracy decreases from ≈ 0.23 to ≈ 0.12 with p and $r = 0.1$, and it decreases from ≈ 0.17 to ≈ 0.09 for



(a) L2-NMF (MUR) on ORL dataset.

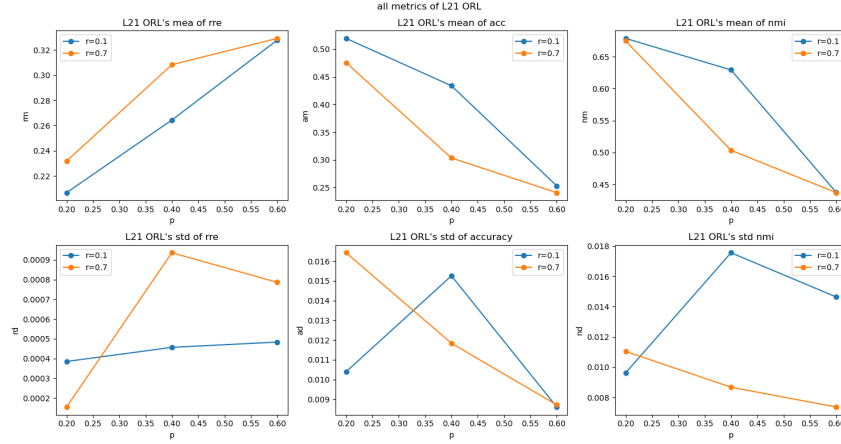


(b) L2-NMF (MUR) on Extended YaleB dataset.

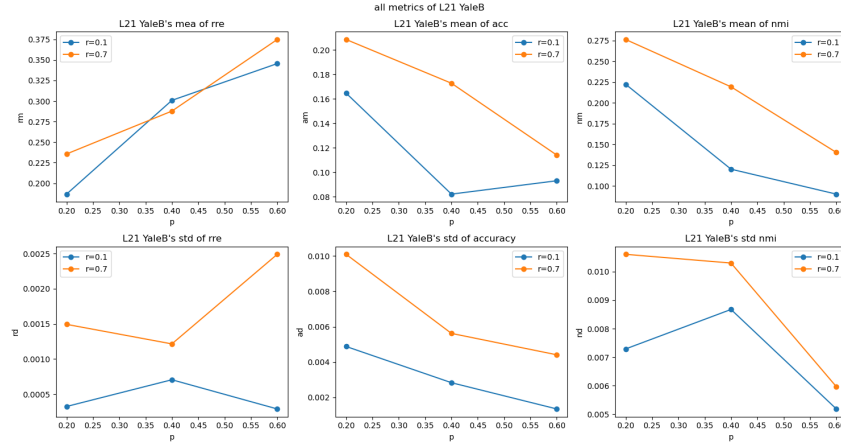
Figure 3: Results of L2-NMF (MUR) under salt-and-pepper noise on ORL (left block) and Extended YaleB (right block). Top row shows the means, bottom row the standard deviations over repeated runs for RRE (left column), Accuracy (middle), and NMI (right) as the noise level varies. $p \in \{0.2, 0.4, 0.6\}$; curves correspond to $r = 0.1$ and $r = 0.7$. The x-axis is the noise level; the y-axis is the corresponding mean or std; **blue** = $r = 0.1$, **orange** = $r = 0.7$.

$r = 0.7$. NMI decreases from ≈ 0.275 to ≈ 0.175 with p and $r = 0.1$ and decreases from ≈ 0.260 to ≈ 0.075 for $r = 0.7$. The values of standard deviations of three matrices are small, indicating stable estimates across runs. However, the values for mean accuracy and NMI are smaller than ORL's, it represents the YaleB dataset is challenging illumination. Overall, the L2,1 demonstrates a moderate RRE, accuracy and NMI with increasing p and r , especially on ORL demonstrating good robustness. Although clustering scores are low on YaleB, L2,1 still limits reconstruction error under heavy noise.

By comparing the MUR algorithm's results, we can clearly see that the performance of mean RRE in both ORL and YaleB is better than MUR. The slope of RRE of MUR is larger as p increases from 0.2 to 0.6, while L2,1 algorithms grow more slowly to achieve lower reconstruction errors at the same (p, r) . In addition, L2,1 has a smaller RRE standard deviation and more stable performance. Furthermore, we notice that the mean Accuracy and NMI values of L2,1-NMF on the ORL dataset are consistently higher than those of MUR as p and r increase. Besides, YaleB has an overall lower accuracy and NMI values for both methods due to more complex lighting. Nevertheless, L2,1 can still control RRE at higher noise and the general values of accuracy and NMI are higher than the MUR algorithm. This indicates that it is more robust in pixel-level recovery. Hence, the L2,1



(a) L2,1-NMF on ORL dataset.



(b) L2,1-NMF on Extended YaleB dataset.

Figure 4: Results for L2,1-NMF under salt-and-pepper noise on ORL (left block) and Extended YaleB (right block). Layout and axes match Figure 2.

NMF algorithm is more robust to sparse large-magnitude noise (salt and pepper) than the L2-NMF algorithm.

To compare robustness briefly, we also include heatmaps of metric means and standard deviations over the salt-and-pepper noises for both algorithms on both datasets. The top two blocks of heatmaps correspond to the L2-NMF (MUR) algorithm, and the bottom two blocks correspond to the L2,1-NMF algorithm. The x-axis represents $r \in \{0.1, 0.7\}$, and the y-axis represents $p \in \{0.2, 0.4, 0.6\}$. The layout is similar to the scatterplots above: in each dataset, the top row shows the means and the bottom row shows the standard deviations over repeated runs for RRE (left column), Accuracy (middle), and NMI (right). Darker colors indicate higher metric values. The heatmaps reinforce the scatter plot trends: L2,1-NMF demonstrates consistently milder RRE increases and smaller variances than MUR. On ORL, it also achieves higher mean Accuracy and NMI, while on Extended YaleB the differences are mixed across (p, r) . Therefore, it reinforces that the L2,1 NMF algorithm is more robust to sparse large-magnitude noise (salt and pepper) than the L2-NMF algorithm.

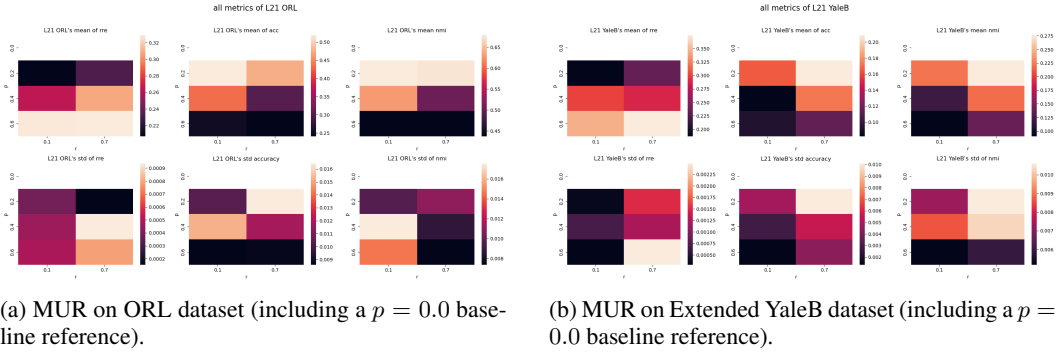


Figure 5: Heatmaps of mean and standard deviation for RRE, Accuracy, and NMI under different (p, r) noise settings for the L_2 -NMF (MUR) algorithm on ORL (left) and Extended YaleB (right) datasets.

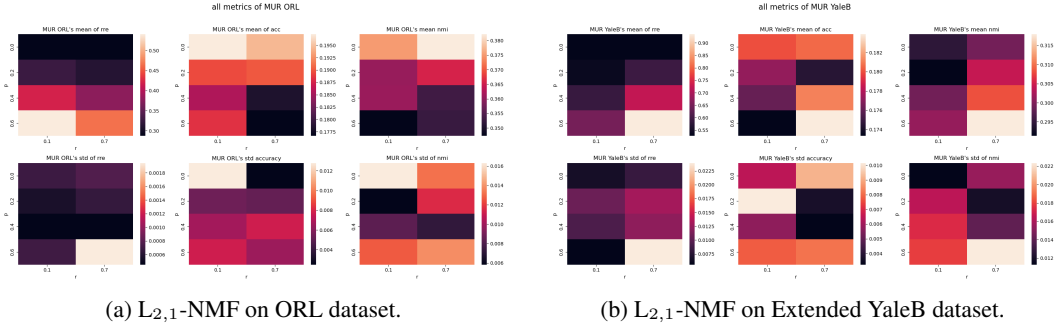


Figure 6: Heatmaps of mean and standard deviation for RRE, Accuracy, and NMI under different (p, r) noise settings for the $L_{2,1}$ -NMF algorithm on ORL (left) and Extended YaleB (right) datasets.

In addition to the heatmaps, we further present bubble and scatter plots to visualize the relationships among the evaluation metrics and reinforce the same trends. The top blocks correspond to the L_2 -NMF (MUR) algorithm, and the bottom blocks correspond to the $L_{2,1}$ -NMF algorithm. For each algorithm, the first row of bubble charts represents results on the ORL dataset, and the second row represents results on the YaleB dataset. The bubble charts combine the means and standard deviations of Accuracy (left column), RRE (middle) and NMI (right) under the same (p, r) settings. The lighter colour and larger the bubble represents higher the values of the corresponding metric. We can clearly see that $L_{2,1}$ -NMF shows smaller and more concentrated bubbles in RRE and lighter and larger bubbles in accuracy and NMI. This reinforces that $L_{2,1}$ -NMF achieves greater robustness under increasing salt-and-pepper noise than MUR.

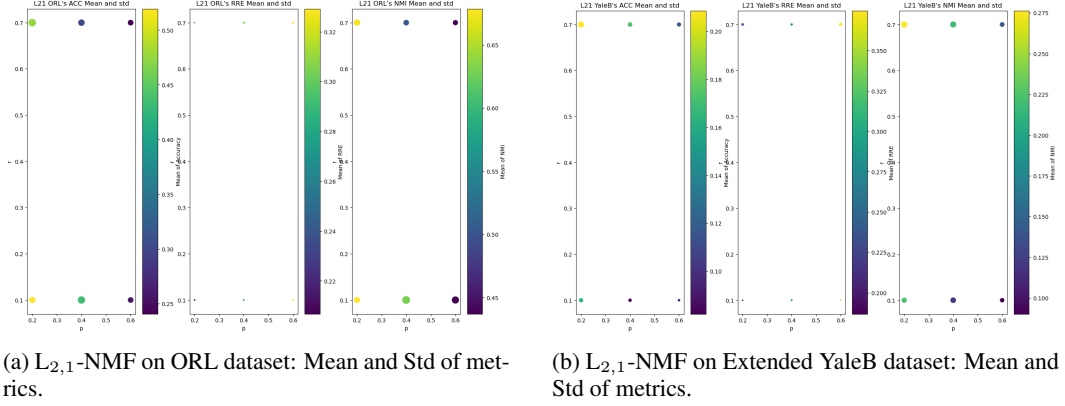


Figure 7: Bubble plots of the mean and standard deviation of Accuracy, RRE, and NMI under different (p, r) combinations for the $L_{2,1}$ -NMF algorithm on ORL (left) and Extended YaleB (right) datasets.

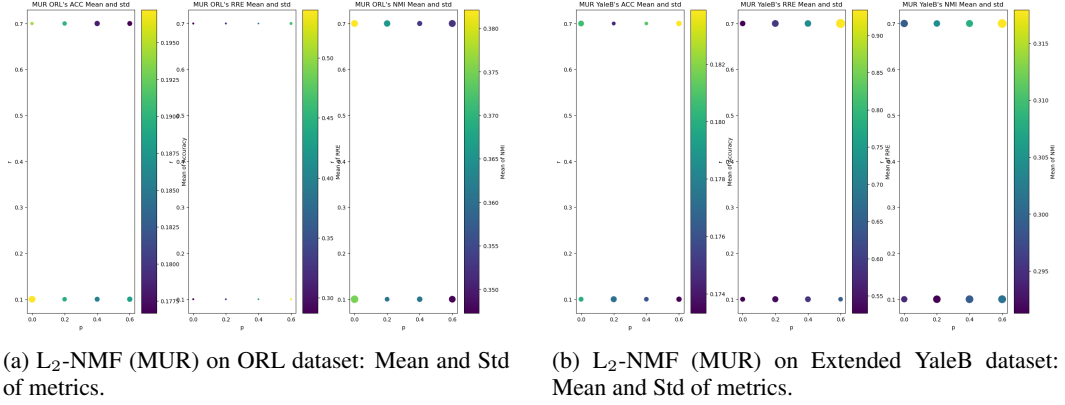
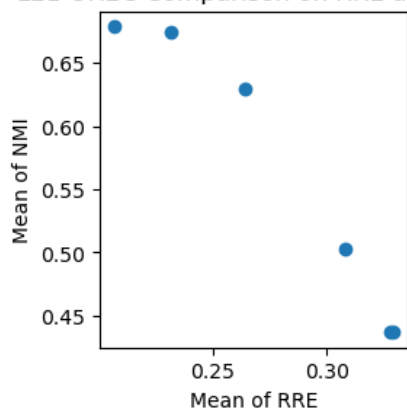


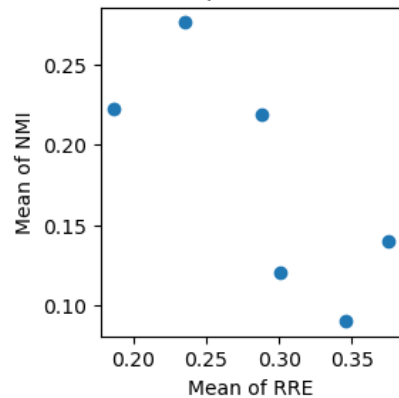
Figure 8: Bubble plots of the mean and standard deviation of Accuracy, RRE, and NMI under different (p, r) combinations for the L_2 -NMF (MUR) algorithm on ORL (left) and Extended YaleB (right) datasets.

Besides, we created additional scatter plots between RRE and NMI (left column), and between Accuracy and NMI (right column), to further confirm consistent metric relationships across both datasets. In general, we notice that there is negative correlation between RRE and NMI, and there is positive correlation between accuracy and NMI across two algorithms and datasets. It indicates that the better reconstruction quality is associated with higher clustering accuracy and stronger label consistency. However, these correlations are less apparent in the YaleB dataset, which reinforces that the YaleB dataset contains illumination variation to disrupt the relationships among metrics. Overall, these relationships further support that the $L_{2,1}$ -NMF algorithm preserves label structure more effectively under increasing noise.

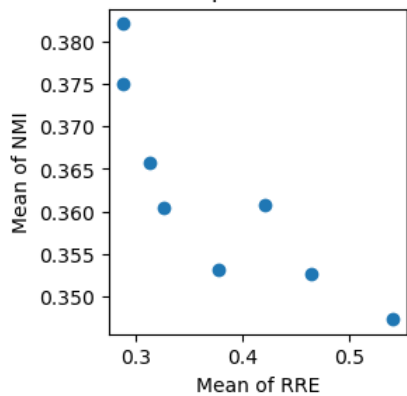
L21 ORL's Comparison on RRE and NMI



L21 YaleB's Comparison on RRE and NMI



MUR ORL's Comparison on RRE and NMI



MUR YaleB's Comparison on RRE and NMI

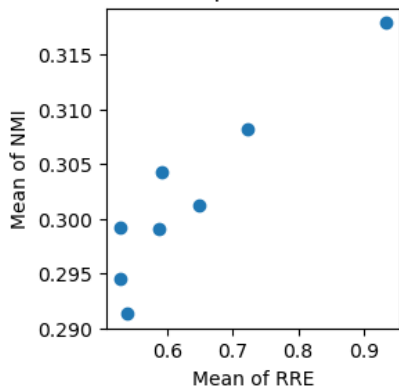


Figure 9: Scatter plots on RRE and NMI

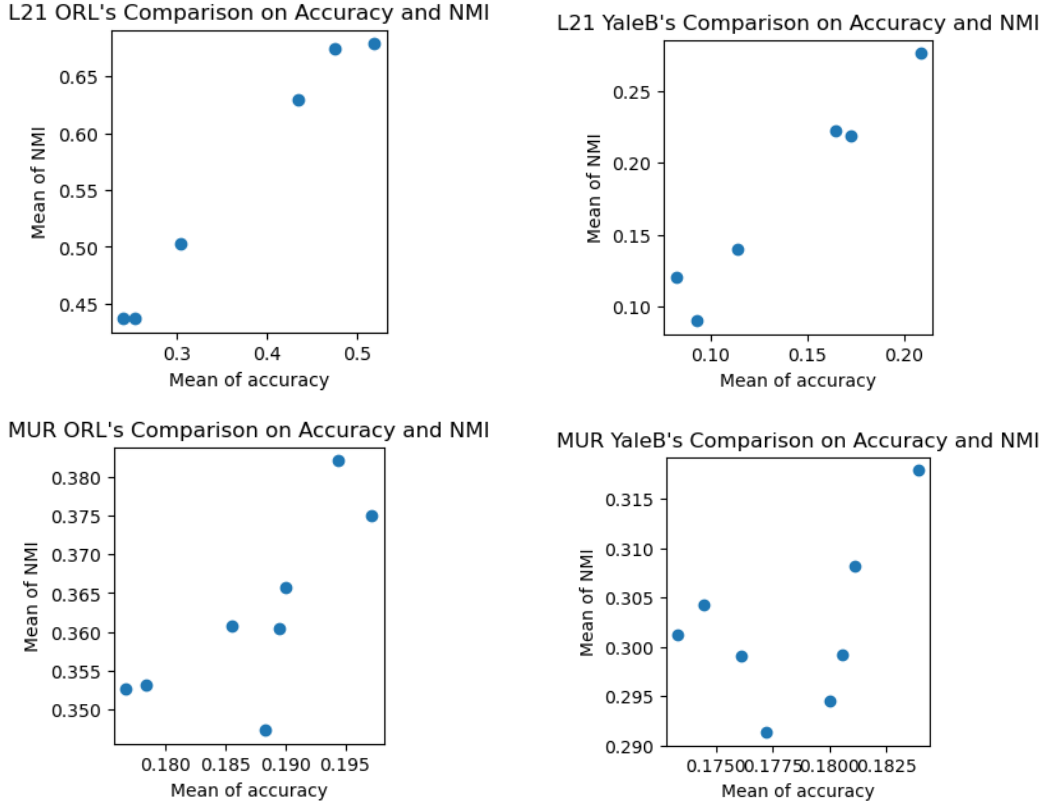


Figure 10: Scatter plots on Accuracy and NMI

In general, these results from visualisation provide complementary evidence that L2,1-NMF achieves lower reconstruction errors, smaller standard deviation, and stronger consistency between clustering metrics. It further supports that L2,1-NMF has stronger robustness against salt-and-pepper noise compared with the standard MUR algorithm.

Overall, these results confirm that L_{2,1}-NMF achieves lower reconstruction errors and more stable clustering under impulsive noise, especially on the simpler ORL dataset, while its advantage becomes less pronounced on the more complex YaleB dataset due to illumination variation.

4.6 Personal Reflection

Throughout the experiment, we learned that implementing robust NMF models requires careful control of initialization, normalization, and numerical stability. The most challenging part was ensuring fairness in comparison — particularly, aligning the rank and iteration settings between L2-NMF and L2,1-NMF so that differences truly reflected algorithmic robustness rather than parameter effects. Visualizing the results helped me understand how salt-and-pepper noise distorts feature distributions and why the L2,1-NMF’s column-wise weighting mitigates this distortion effectively. In hindsight, we realized that deeper parameter tuning or additional noise types (e.g., Gaussian or occlusion noise) could further strengthen the experimental validation. Overall, this project enhanced my understanding of how robust matrix factorization techniques behave under impulsive noise and how evaluation metrics reveal their stability.

5 Conclusions and Future Work

Overall, we researched on the robustness of two NMF algorithms, L2-NMF (MUR) and L2,1-NMF, under salt-and-pepper noise on face datasets of ORL and Extended YaleB. After standardized pre-processing and implementation of algorithm and evaluation metric, we evaluated performance by

Relative Reconstruction Error (RRE), Average Accuracy, and NMI, and demonstrated them by scatter plots and heatmaps. Across all (p, r) settings, L2,1-NMF consistently has lower RRE than MUR on both datasets, and it generally shows smaller RRE standard deviations. Moreover, it also achieved higher accuracy and NMI on the ORL dataset than MUR at the same (p, r) settings. On Extended YaleB, the mean accuracy and NMI are low for both methods due to illumination changes, and the differences between two methods are sometimes mixed. However, L2,1 still limited reconstruction degradation at high levels of noise, indicating stronger robustness. The heatmaps summarized the full grid and reinforce the same trend, which is that L2,1's RRE grows more slowly with p and stays more stable, especially when $r = 0.7$. Overall, the evaluation results support that L2,1-NMF is more robust to sparse, large-magnitude impulse noises than the L2-loss MUR baseline. These gains are most obvious on ORL, while on YaleB the clustering advantage is not always uniform.

For future work, we will focus on handling YaleB's challenging illumination. We plan to apply histogram equalization to enhance the image contrast by altering the intensity distribution of the histogram [10]. It can stretch the grayscale distribution to make the dark areas brighter and the bright areas less overexposed at the preprocessing step. In this way, it may improve the clustering performance and stability on YaleB dataset. Besides, we also plan to test on larger and more complex datasets to further study the robustness of the two algorithms in different noise and structure conditions.

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